智能纪要：Discussion on IEEE paper 2026年2月5日

> 会议主题：Discussion on IEEE paper
>
> 会议时间：2月5号（周四） 16:31 - 17:56 （GMT+08）
>
> 参会人： Bowen QING Enge Lou Haoxuan Jia Shifana RAHMAN Yunuo Lin

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总结

Bowen QING 10510395 2026-04-19 10:21 UTC

AI-Powered Functional Safety for Driver Monitoring Systems

Key Research Focus

L1 Baseline System Requires Upgrade

Current YOLO implementation needs replacement with state-of-the-art open source models

- Focus on facial features and eye closure detection
- Higher frequency decision making
- Faster response time but lower accuracy
- Requires open source SOTA model as baseline

L2 LLM Checker In Development

Explainable AI approach using chain-of-thought reasoning for validation

- Cross-validate L1 outputs with reasoning steps
- Utilizes both DMS and OMS camera inputs
- Higher accuracy but slower response (4s latency)
- Focus on body posture and comprehensive analysis

Safety Framework Established

Multi-level escalation protocol for driver alertness management

- 4-second reaction time requirement
- Progressive warnings: visual → vibration → emergency stop
- Risk mitigation through redundant systems
- Functional safety levels (A-D) with D being most stringent

Research Implementation Plan

Task Category	Specific Requirements	Responsible	Status
Baseline Research	Identify and evaluate open source SOTA models for L1	Heng	In Progress
Dataset Preparation	Label and prepare diverse dataset for training	Jenny & Hao Xuan	Pending
L2 Implementation	Develop chain-of-thought validation framework	Bowen	In Progress
Performance Analysis	Latency and computational overhead evaluation	Team	Pending

Critical Next Step: Replace current YOLO implementation with open source SOTA models as baseline, then implement L2 LLM checker using explainable AI approach with chain-of-thought reasoning to validate driver state detection accuracy within 4-second latency requirements.

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The meeting focused on functional safety in vehicle systems, especially the application of AI in the Driver Monitoring System (DMS). It discussed ways to ensure the safety of electrical and electronic systems, the use of L1 and L2 models, and how to address review comments on a related paper. The content is as follows:

- **Functional Safety Basics**
  - **Concept of Functional Safety**
    - Shifana RAHMAN explained that functional safety is about dealing with failures in a vehicle's electrical and electronic systems. Since software and hardware can fail, it is necessary to analyze potential scenarios and reduce risks. For example, airbag deployment is a functional safety mechanism. Analyses are conducted on various situations such as late or early deployment and their impact on drivers.

        - Different levels (A, B, C, D) of functional safety are defined, with D being the most stringent. Functional safety aims to mitigate risks rather than completely avoid accidents.
    - **Redundancy Mechanisms**: To enhance safety, redundant components are often used. For instance, if one sensor fails, another can take over. In the case of pedestrian detection, multiple sensors like cameras, radars, or ultrasonic sensors can be used. If a customer has a limited budget, they may choose a more cost - effective combination.
- **AI Safety in DMS**
    - **Ensuring AI Correctness**
        - Shifana RAHMAN pointed out that when using AI algorithms in the DMS, there is a risk of incorrect outputs, similar to the hallucination problem in LLMs. To address this, the concept of Explainable AI (XAI) is introduced.
        - By using a chain of thought, similar to the reasoning module in ChatGPT, the steps taken by the AI are recorded and evaluated. The output is cross - checked with the input to ensure that the AI is performing correctly. For example, if the AI calculates “a + b = c”, it is checked whether “c - b = a” and whether the values of “a” and “b” are correctly used.
    - **DMS System Improvement**
        - In the DMS system, which consists of a camera, a DMS algorithm, and an output of the driver's state, safety can be enhanced by adding redundant components. For example, an additional camera can be added in case the first one fails.
        - For the DMS algorithm, a duplicate backup operator is not sufficient. Instead, a fast algorithm and a more complex one can be used for cross - validation. The DMS can be used as the main input, and the OMS as a secondary input to increase safety.
- **L1 and L2 Model Usage**
    - **Model Characteristics**
        - The L1 model, such as the self - designed Yolo model, provides fast feedback but has lower accuracy. The L2 model, which is an LLN, has higher accuracy but is slower and may have latency.
        - L2 can perform offline checks on L1's output. It takes a sample of L1's output sequence for evaluation, which is not synchronized with L1.
    - **Combined Model Application**

- The two models should be used in combination. When L1 and L2 give different conclusions, L2 is generally believed, except in the third case where L1 is trusted.
- To ensure the safety of passengers, a buffer time of up to 4 seconds is acceptable. For example, when detecting driver drowsiness, the system should be able to react within this time frame. If the driver is detected as sleepy, a series of escalation warnings can be issued, and if necessary, an emergency stop can be carried out.

Driver Monitoring System

Case	Actual Driver State in video (Ground Truth)	LLM Decision	Interpretation
True Positive (TP)	Unsafe (drowsy/distracted)	Unsafe	Correct detection of safety risk
True Negative (TN)	Safe (alert)	Safe	Correct rejection of false alarm
False Positive (FP)	Safe (alert)	Unsafe	False alarm
False Negative (FN)	Unsafe (drowsy/distracted)	Safe	**Hazardous miss** (most critical)

Driver Monitoring System Detection Cases(shared content)

- **Data Set and Model Selection**
  - **Use Case Focus**: The team will focus on the driver BP use case. The case of children extending their bodies outside the window is considered not a functional safety case but a passive safety case, as it does not directly affect driving functionality.
  - **Model Baseline and Tuning**
    - Open - source models will be used as baselines. The team will deploy these models locally, retrain them on their own data sets, and fine - tune the models to improve accuracy.
    - The use of a behavior tree in the prompt for the LLM is proposed. By adding more context to the prompt, the behavior of the L2 model can be improved.
- **Handling Review Comments**
  - **Assignment of Tasks**
    - Specific team members are assigned to handle different review comments. For example, E N G will research baselines to address the issue of missing baseline comparisons.
    - Jenny and Hao Xuan will cooperate to prepare the data set, label the data, and train a small model to check if it can improve accuracy.
    - Heng will be responsible for ablation experiments, which are similar to the baseline research in nature.

Reviewer	Category	Review Comment (As Received)	Action Point (Recommended)	Responsible
All	General	Thank you for your submission and for addressing an important and emerging topic in automotive AI safety. The reviewers found the problem domain compelling and appreciated the attempt to structure safety monitoring through a multi-layer CoT-inspired framework, major revision is needed	Plan major revision; restructure paper and experiments to address conceptual, methodological, and experimental gaps	Shifana Rahman
Reviewer 3	Key Issue 1	Fundamental Logical Inconsistency: AI models lack safety guarantees, yet the solution uses a VLM as Level 2 safety monitor	Explicitly acknowledge LLM limitations and reposition L2 as a risk-reduction monitor, not a safety-certified component	Shifana Rahman
Reviewer 3	Key Issue 1	LLM-based monitor not justified as more trustworthy than Level-1 perception model	Add a subsection explaining L2 as an independent, diverse checker with constrained prompts, bounded outputs, and runtime safeguards	Shifana Rahman
All	Key Issue 1	Must justify VLM determinism/verifiability or propose mitigation strategies	Add mitigation strategies: prompt freezing, temperature=0, output schema constraints, confidence thresholds, fallback logic	Shifana Rahman
Reviewer 3	Key Issue 2	Conceptual ambiguity and misuse of "Chain-of-Thought"	Rename Method 1 to "Structured Semantic Validation" or "Deterministic Reasoning Chain"	Shifana Rahman
Reviewer 3	Key Issue 2	Method 1 is conventional multi-stage validation logic, not genuine CoT	Clearly distinguish deterministic reasoning from LLM-based CoT; avoid calling Method 1 "CoT"	Shifana Rahman
All	Key Issue 2	Recommendation to distinguish or genuinely implement CoT	Keep Method 1 deterministic; reserve "CoT" terminology strictly for LLM-based reasoning in Method 2	Shifana Rahman
Reviewer 1	Key Issue 3	Missing baseline comparisons	Add baselines: perception-only, rule-based checker, threshold-based monitor	Bowen

RASIC Table for Review Comment Assignments(shared content)

- **Other Review Comment Solutions**
  - The team will use only one method in the paper to avoid complexity. They will also explicitly state video length, frame count, and decision windows in the data set description.
  - For the issue of data set diversity, the team will try to make the data set more diverse. A pseudo - code and a flow - chart will be created to address the abstract Cod implementation.
  - The scalability and computational overhead analysis will be carried out using a personal PC to simulate on - vehicle hardware. The team will also try to find out - of - distribution cases in their data set but may not conduct a full - scale OOD evaluation due to time constraints.

- **Other Discussions**
  - **Paper Submission and Group**
    - A new group will be created for the paper submission to avoid Huizhou guys knowing about it. Shifana RAHMAN will send an Excel sheet to the group.
    - The paper will be written in LaTeX, and team members need to provide their Gmail IDs for access.
  - **Internship and Leave**
    - Although there are no specific points or rewards in school for publishing the paper, it can be added to the resume when applying for jobs.
    - The team discussed Chinese New Year leave, and Shifana RAHMAN will catch up with the team later. The paper needs to be submitted before March 1st.

AI notes feedback collection [Export is restricted for this type of content.]

智能章节

00:01 **Discussion on L1 and L2 Model Latency, Synchronization, and Buffer Time for Safety Assurance**

> Shifana RAHMAN discusses using Sota and the relationship between L1 and L2. L2 is a flow model (LLN) that checks L1 offline, taking a few samples from L1's output with a slight delay. A buffer time is considered to ensure passenger safety, with a suggested buffer of about 4 seconds as per a reviewer's mention. In the scenario of detecting driver sleepiness, a longer buffer time can be tolerated to prevent accidents within 4 seconds.

03:00 **Discussion on Reaction Time and Mitigation Methods of Driver Monitoring System**

> Shifana RAHMAN discusses the reaction time for the driver monitoring system, suggesting it should be within 4 seconds and needing cross - checking. She mentions gentle wake - up calls, multiple escalation warnings, and an emergency stop if the driver doesn't wake up. Mitigation methods involve safely parking the car. She also proposes researching current L1 systems as a baseline, aiming for better performance. Additionally, she notes that their VLM may detect high - level behaviors like head - nodding, which current L1 systems can't.

05:29 **Discussion on Placing Level 2 on Level 1 and Search for Reasons**

> Shifana RAHMAN addresses a reviewer's question about why level 2 isn't placed on level 1. She explains that L2 requires more time, has latency but higher accuracy, while L1 offers faster feedback. She'll note this when writing. Regarding the customer, the payment remains the same due to extra costs for a checker and its module development. She also plans to search for more reasons, perhaps using ChatGPT.

06:38 **Classifying Datasets for Combined Use of Low and High-Accuracy Models**

> Shifana RAHMAN suggests using the chain of thoughts from ChatGPT's reasoning without its low model, instead using a low - level, faster model. She proposes classifying datasets into high - risk and low - risk groups. High - risk data needs a fast model, while low - risk data can use more time for higher accuracy. By comparing with L1 (faster, lower accuracy) and L2 (slower, higher accuracy), the two can be combined to achieve higher accuracy and speed in the experiment.

08:47 **Discussion on Cross - checking the Output of Classification Model and Case Exclusion**

> Shifana RAHMAN questions whether feedback from level 2 should be considered instead of level 1. She doubts the necessity of certain input - output in a black - box model, as it's hard to know what happens inside. She explains cross - checking the output with scenarios, like checking if a car stops at a red light. For a classification model with driver's data output, she suggests re - checking high - risk cases and focusing on safe cases, and also mentions that the third case is not needed in the paper.

12:07 **Discussion on Resolving Conflicts between L1 and L2 in Driver Sleepiness Detection**

> Shifana RAHMAN suggests removing the circuit due to potential conflicts between L1 and L2. She emphasizes the importance of ensuring driver safety. In normal cases, L1 should correctly detect a driver's sleep state. L2 is used as a checker for L1. When L1 and L2 conflict, L2 is generally trusted, except in the third case where L1 is believed. For the fourth case, L2 is used, and it needs to be made more reliable.

17:37 **Discussion on Using DMS and OMS Cameras for L1 and L2 to Enhance Driver Monitoring Reliability**

> Shifana RAHMAN suggests L1 and L2 should focus on the camera, with L1 focusing on the driver's facial features and having a higher decision frequency, while L2 focuses on the big picture including body posture and movements with a lower decision frequency but higher accuracy. She thinks using both DMS and OMM cameras can make L2 more reliable. She also mentions the difficulty of accurately detecting driver sleepiness and asks to see sample video data.

22:09 **Advantages of L1 and L2 in Hazard Detection and Alarm Cancellation**

> Shifana RAHMAN points out that the advantage of L1 is it can detect hazard situations faster than L2. For instance, L1 can act immediately when a DP occurs, while L2 may take about three seconds to detect a sleepy situation. L2 can cancel the alarm after cross - validation if it finds L1's alarm is false. In a true dangerous case, L2 will keep alarming the driver. To make L2 more accurate, more information like car status should be added.

23:38 **Driver Monitoring and Functional Safety Discussion: Alarms, Use Cases, and Research Directions**

> Shifana RAHMAN discussed vehicle safety features. She argued alarms should not be manually canceled to avoid danger. Multiple warnings should precede taking vehicle control or parking. She also mentioned creating an architecture for driver monitoring, researching related work. The focus is on driver BP use - case. She distinguished functional and passive safety, saying children extending bodies is passive, while driver monitoring is functional, aiming to reduce risk.

35:55 **Discussion on the Differences between Functional Safety and Active Safety and Classification Criteria**

> Shifana RAHMAN discusses the difference between functional safety and active safety, suggesting active safety falls under functional safety. She differentiates passive and functional safety based on when an action or alarm occurs. To classify safety, she mentions considering accident severity, exposure (probability of the event), and controllability. She notes OEMs haven't classified children putting hands outside as functional safety, while DMS is, highlighting ensuring driver control is key.

40:25 **Discussion on Using the First Case for Research and Related Strategies**

> Shifana RAHMAN suggests using the first case as there are few papers and datasets on it, offering a higher chance of novelty. She points out issues like lack of related work and baselines. Time limitation is also a concern when changing datasets. She proposes using a large knowledge model, labeling data, and using Gemini or Chat GPT API for beta - testing. Finally, she seems to suggest deciding on using this case.

44:51 **Discussion on Baseline Comparisons and Dataset Preparation for Research**

> Shifana RAHMAN addresses previous questions. For missing baseline comparisons, research is needed to determine which baselines to take, and E will conduct the research review. Regarding evaluation limited to a single proprietary use case, the same data sets should be used. Before deploying the open - source model locally, GPK should be used to label the data set. Training the data set may take a weekend due to slow network speed, and 100 high - quality data are needed for fine - tuning.

48:45 **Discussion on Simplifying Evaluation with Single Method and Baseline for Proprietary Use Case**

> Shifana RAHMAN points out that the experimental evaluation is limited to a single proprietary use case, making it hard to assess if Cod truly improves safety, performance, or robustness. She decides to use only one method instead of two to avoid complexity. Regarding baselines, she plans to remove most of the written ones, keeping only one. Ablation experiments are also mentioned, and she thinks Heng should be responsible for deciding which baseline to use.

52:26 **Discussion on Dataset, Model Configuration, and Cooperative Task of Data Labeling and Model Training**

> Shifana RAHMAN points out that data set description, model configuration, and latency determinism are only briefly treated. She asks about who will handle the data sets. She suggests cooperation with Jenny on a task that involves labeling data, training a small model, and using it as L2 to improve accuracy. She also mentions Hao Xuan and asks them to do something, though the details are a bit unclear.

54:10 **Discussion on Latency, Throughput, Computational Cost Analysis, Dataset Diversity and Video Labeling**

> Shifana RAHMAN raises several points in the meeting. She asks about who will assist with latency, throughput, and computational cost analysis. For video data, she emphasizes explicitly stating video length, frame count, and decision windows. She'll check if 3 - 4 seconds is acceptable in the real - world and provide latency time. Also, she points out that the dataset is not diverse enough and assigns two people to handle this issue related to previous experiments.

56:08 **Discussion on COD Implementation Abstract, Pseudo - code, Flow - chart for Small Language Model Training and Visual Open - source Project**

> Shifana RAHMAN discusses the implementation of code, specifically the need for a pseudo - code. She volunteers to do it and agrees that someone should proceed with it. Additionally, she mentions creating a flowchart based on the training process of a smaller language model and covering all scenarios in the code. There is also a mention of opening a purely visual open - source project, though the details about quantity seem incomplete.

57:13 **Analysis of Scalability and Computational Overhead with PC Simulation of Vehicle Chipset and Work Division**

> Shifana RAHMAN discusses analysis, scalability, and computational overhead. Due to the lack of real vehicle hardware, a personal PC will be used to simulate on-vehicle chipsets, and this should be noted in the report. They can check Jackson chips for intelligent driving from Simon. Quantitative analysis is required, including calculating system latency and frequency. The work will be divided among three people.

01:01:21 **Discussion on OOD Evaluation and Mining OD Cases in Dataset**

> Shifana RAHMAN mentions that the three people are dealing with a common issue of missing OOD (Out of Distribution) evaluation. She lists examples like adults, pets, night - time IR, and tinted windows as out - of - distribution cases. She thinks they won't conduct this evaluation due to lack of time but may just mention it. They have many OD cases and will try to mine some OD cases from their dataset, which might include edge or hard cases.

01:03:01 **Discussion on New Paper Group Creation, Submission, and Collaboration Arrangements**

> Shifana RAHMAN plans to write articles first and then share them. She suggests creating a new group to avoid Huizhou colleagues knowing about the paper submission. She'll put an Excel sheet in the group. Publishing papers can be added to resumes. ChatGPT can't be used due to detection. She'll create a new paper and share access via Gmail IDs. The paper has a March 1st deadline, and she'll catch up with others later.

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